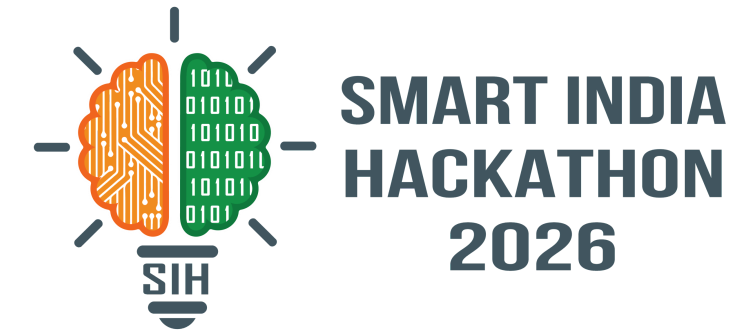
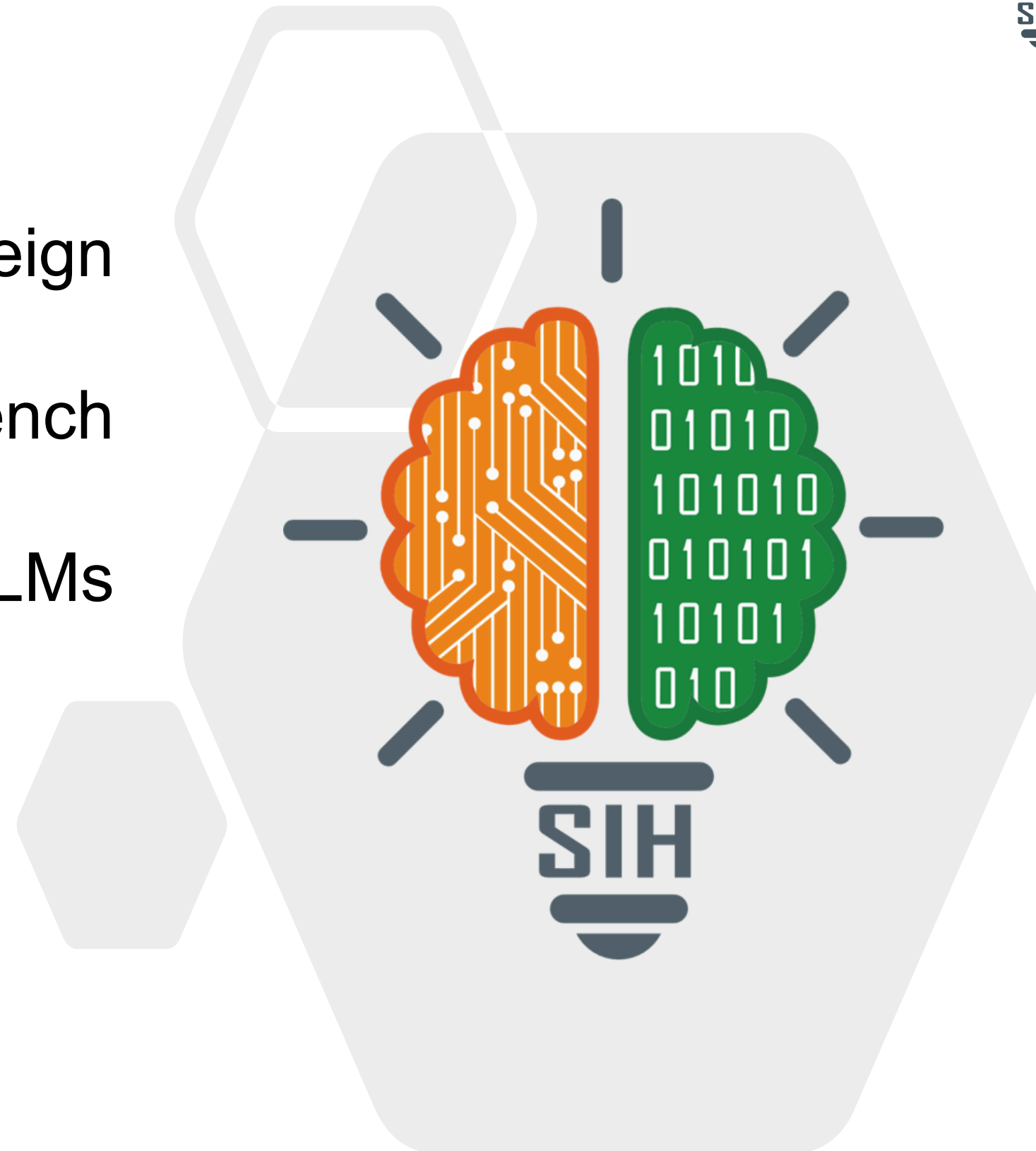


SMART INDIA HACKATHON 2026



- **Problem Statement ID-** 26117
- **Problem Statement Title-** Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team Name-** Orchestra



IDEA/SOLUTION:

A self-hosted, air-gapped AI workbench for confidential industrial work.



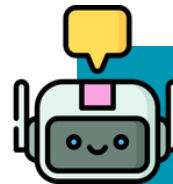
Document Generation

Excel files, PPTs, Word reports all in predefined templates.



Multimodal Analysis

OCR+VLM for scanned PDFs, engineering drawings, handwriting.



Coding Agents

Write code, executes, iterates inside network isolated Sandbox.



Agentic Core

Multi-step planning, automatic tool calling for complex workflows.



Audit Logs

Prints every step of calculations and logs every model, tool call.



Locally Hosted

Local knowledge base grounds answers in organisation's SOPs.

PROBLEM RESOLUTION:

- **Confidentiality maintained** in regulated industries (at scale).
- **On-premise deployment** letting teams use AI despite a **cloud AI ban** in sensitive sectors.
- **Saves engineer time** by **auto-generating structured data from scans**.

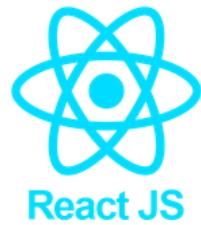
UNIQUE VALUE PROPOSITIONS (UVP):

- **Extensible model:** New models plug in via YAML, no rewrite needed.
- Breaks the task into steps with **4 models** on standby; automatically picks the right model at each step.
- **Parallel document processing** across many documents at once.



TECHNOLOGIES USED

Interface:



TAURI

Backend Serving:



Model Runtime:



LLaMA++

Retrieval & Persistence:



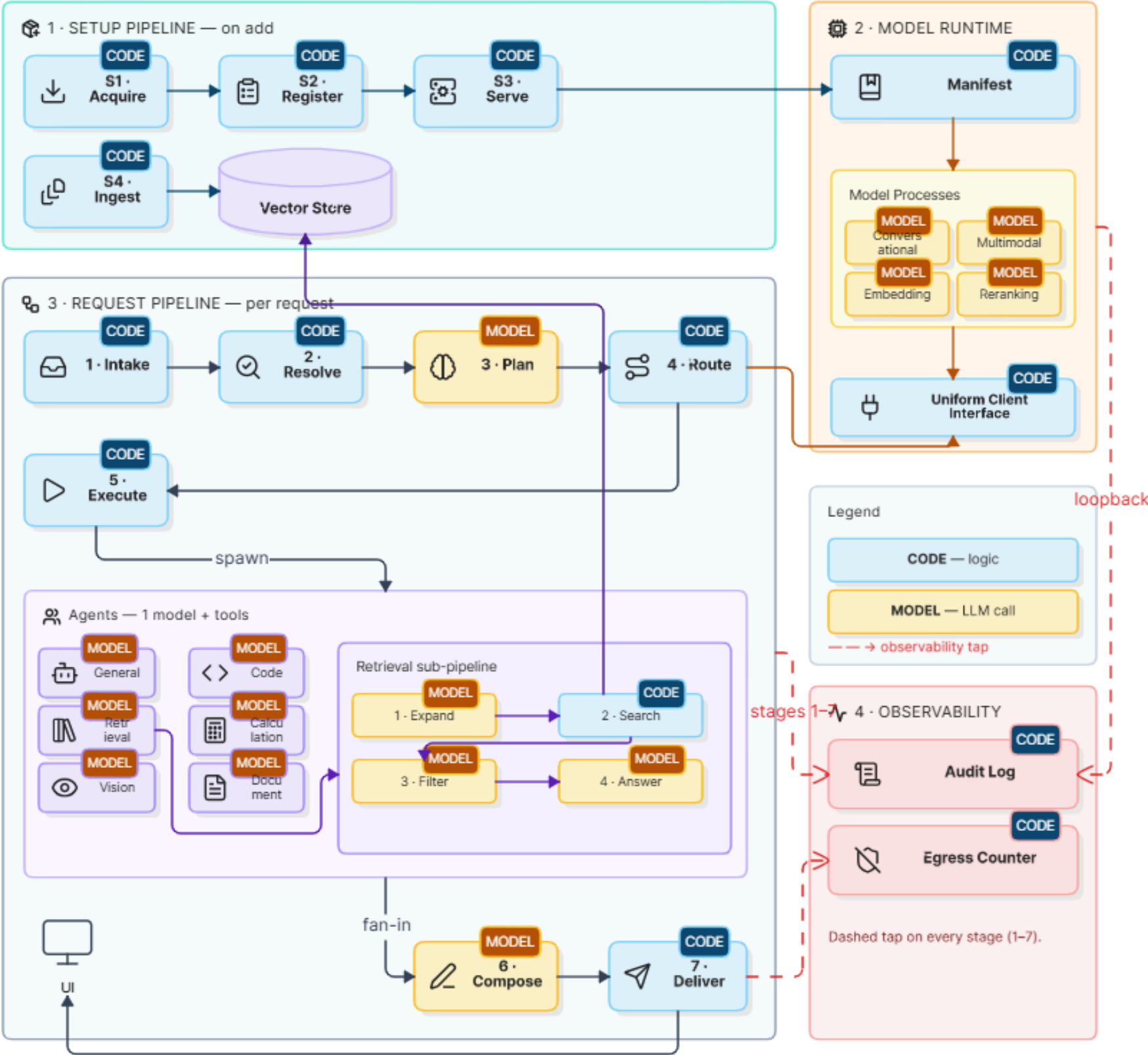
Execution & Distribution:



ARCHITECTURE



SMART INDIA
HACKATHON
2026





*“The unit is not a token.
It is an hour of engineer time.”*

Feasibility & Viability

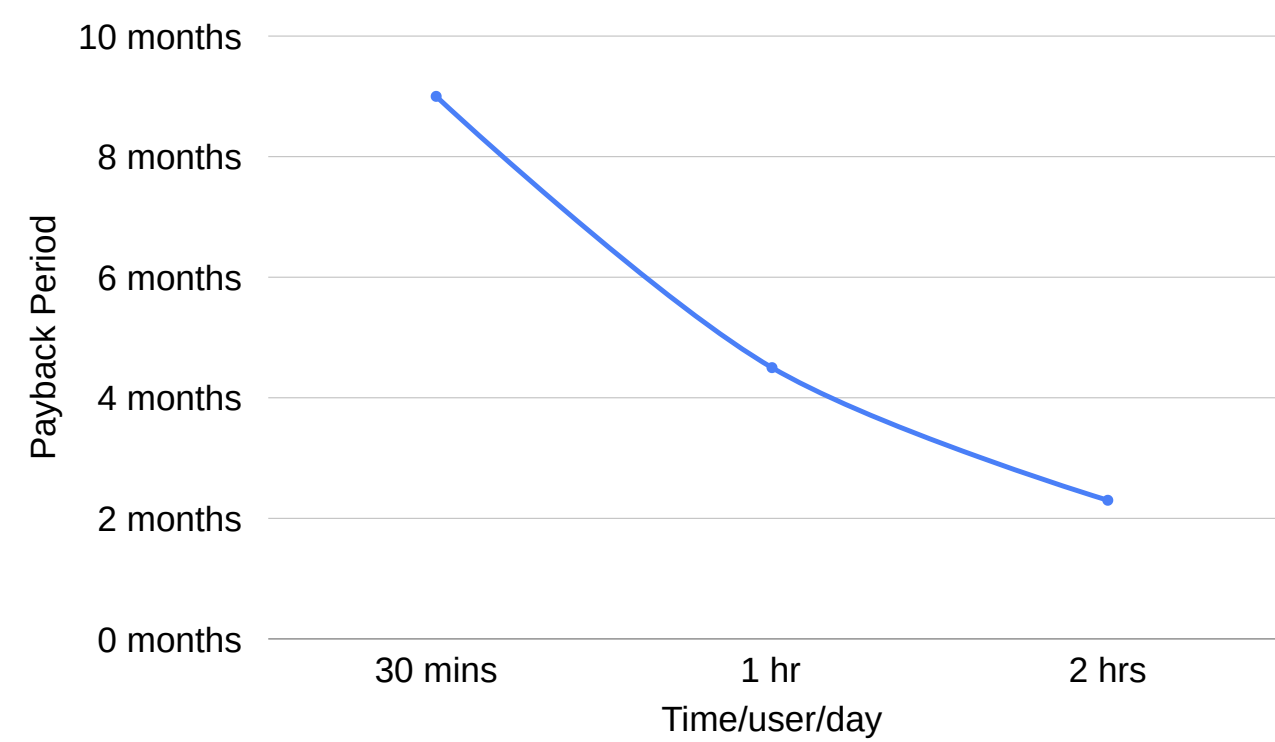
4.5 mo
Payback
period

₹1.25 cr
Value
returned/
yr

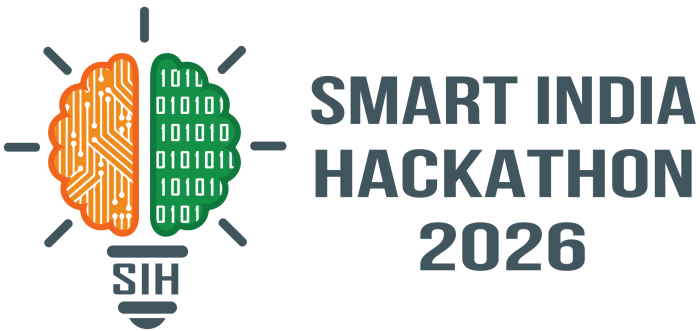
757%
Four-year
ROI

₹110
Cost / hour
returned

When built on a single H100 80 GB server holding four models resident, serving fifty named users across process engineering, inspection, projects and materials.



Risks	Strategy	Status
Small on-premise open weight models less reliable at open ended planning	Fixed roster of 7 agents with the planner emitting one validated plan	✓
Even "offline" tools can leak data via hidden telemetry or license check calls	Sovereignty demonstrated live where logs show zero external calls	✓
Sovereign on-premise hosting is commoditizing fast, with risk of no lasting competitive moat	Shift the moat entirely to the orchestration layer above the model	✗
Needs sustained usage to justify GPU hardware cost	Pilot in one department first, then scale hardware tier once proven	✗



On-premise LLMs are proven in banking & defence, while our edge is intelligent multi-model orchestration.

No per-seat licence. No per-token billing. The cost is flat regardless of how heavily it is used.

All ingested content is treated as untrusted data, and requires human approval before file writes.

Secure automation, real impact on-premise, at scale.



SOCIAL IMPACT:

Removes incentive for shadow IT by keeping confidential data off public AI tools.
Logs in every model and tool action for built-in audit and compliance.



ECONOMIC IMPACT:

Frees up significant engineer time from routine paperwork for higher-value work, thus improving throughput without adding headcount.



ENVIRONMENTAL IMPACT:

One on-prem server reduces redundant cloud compute and repeated manual effort.
Digitizing records cuts reliance on printed documentation, helps lower digital & physical waste.

BENEFITS



Engineers reclaim about 1 hour per day from manual drafting, data extraction, and summarizing correspondence.



Institutional knowledge in SOPs and correspondence becomes searchable with citations, not lost when senior staff retire.



Faster turnaround on approval notes, inspection summaries, and board decks means quicker decision cycles across departments.

Technical References

- Local LLM + RAG deployments — proven in banking, defense, healthcare
- Models: Qwen3-Coder, DeepSeek-OCR (open-weight, benchmarked)
- Infra: Qdrant (vector DB), Docker (sandboxed execution)

LLM Research Areas

- Confidence via multi-agent agreement — active LLM research area;
- arXiv 2602.11619, "Behavioral Consistency as an Uncertainty Signal for LLM Agents"

Confidentiality and Compliance Agreements

- Data residency & compliance regulations for refineries
- NCIIPC critical infrastructure & CERT-In mandatory reporting norms (energy sector)